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NVIDIA NeMo Guardrails

5–6 minutes

NVIDIA NeMo Guardrails for Developers

NVIDIA NeMo™ Guardrails is a scalable solution for orchestrating AI guardrails that keep agentic AI applications safe, reliable, and aligned. It allows you to define, orchestrate, and enforce guardrails for content safety, topic control, PII detection, RAG grounding, and jailbreak prevention—all with low latency and seamless integration. Extensible and customizable, it integrates with frameworks like LangChain, LangGraph, and LlamaIndex, supports multi-agent deployments, and leverages GPU acceleration for low-latency performance. NeMo Guardrails includes out-of-the-box NVIDIA Nemotron models packaged as NVIDIA NIM™ microservices and on Hugging Face—covering content safety, topic control, and jailbreak detection—alongside a growing ecosystem of AI safety models, rails, and observability tools. It's part of the larger NVIDIA NeMo software suite for building, monitoring, and optimizing AI agents across their lifecycle.

Access SDKTry MicroserviceDocumentation

See NVIDIA NeMo Guardrails in Action

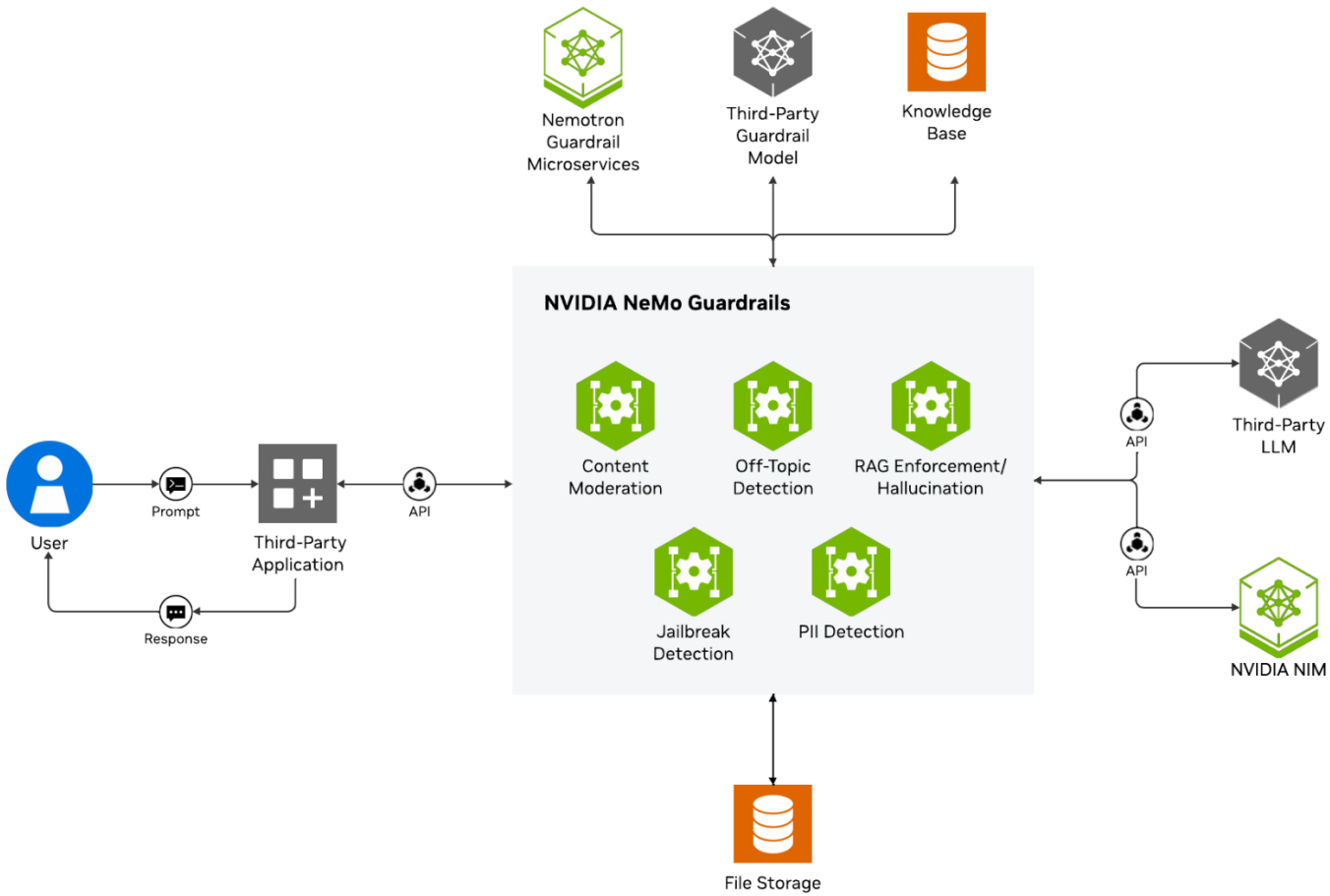
Enforce content safety, RAG grounding, and jailbreak prevention while building secure, compliant AI agents. This video demonstrates how NeMo Guardrails streamlines guardrail orchestration for safer, more reliable AI applications.

How NVIDIA NeMo Guardrails Works

NeMo Guardrails provides components for building a robust, scalable guardrail solution for LLM applications and agents. It evaluates user inputs and model responses based on use-case-specific policies, providing an additional layer of safeguards beyond what's natively available.

Key Benefits:

- **Programmable Policies:** Supports customizable content moderation, PII detection, topic relevance, and jailbreak detection tailored to your industry and use case.
- **Effective Orchestration:** Screens both user inputs and model outputs, effectively orchestrates multiple rails with the lowest latency.
- **Enterprise-Grade Support and Scale:** Handle high volume and scale to multiple applications with enterprise-grade support.
- **Flow Management:** Block, filter, or tailor next action or responses based on your requirements with flexible actions.



Introductory Blog

Simplify building trustworthy LLM apps with AI guardrails for safety, security, and control.

Deploy Guardrails Tutorial

Run Inference with Parallel Rails using NeMo Guardrails microservice.

Example Configurations

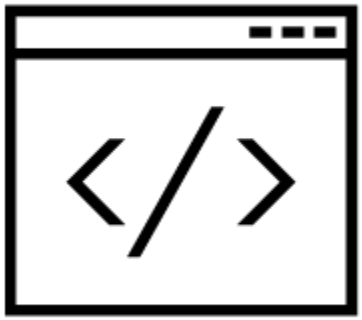
The configurations in this folder showcase various features of NeMo Guardrails, e.g., using a specific LLM, enabling streaming, and enabling fact-checking.

Customer Assistant Example

Learn how to integrate advanced content moderation, jailbreak detection, and topic control with NeMo Guardrails microservices.

Ways to Get Started With NVIDIA NeMo Guardrails

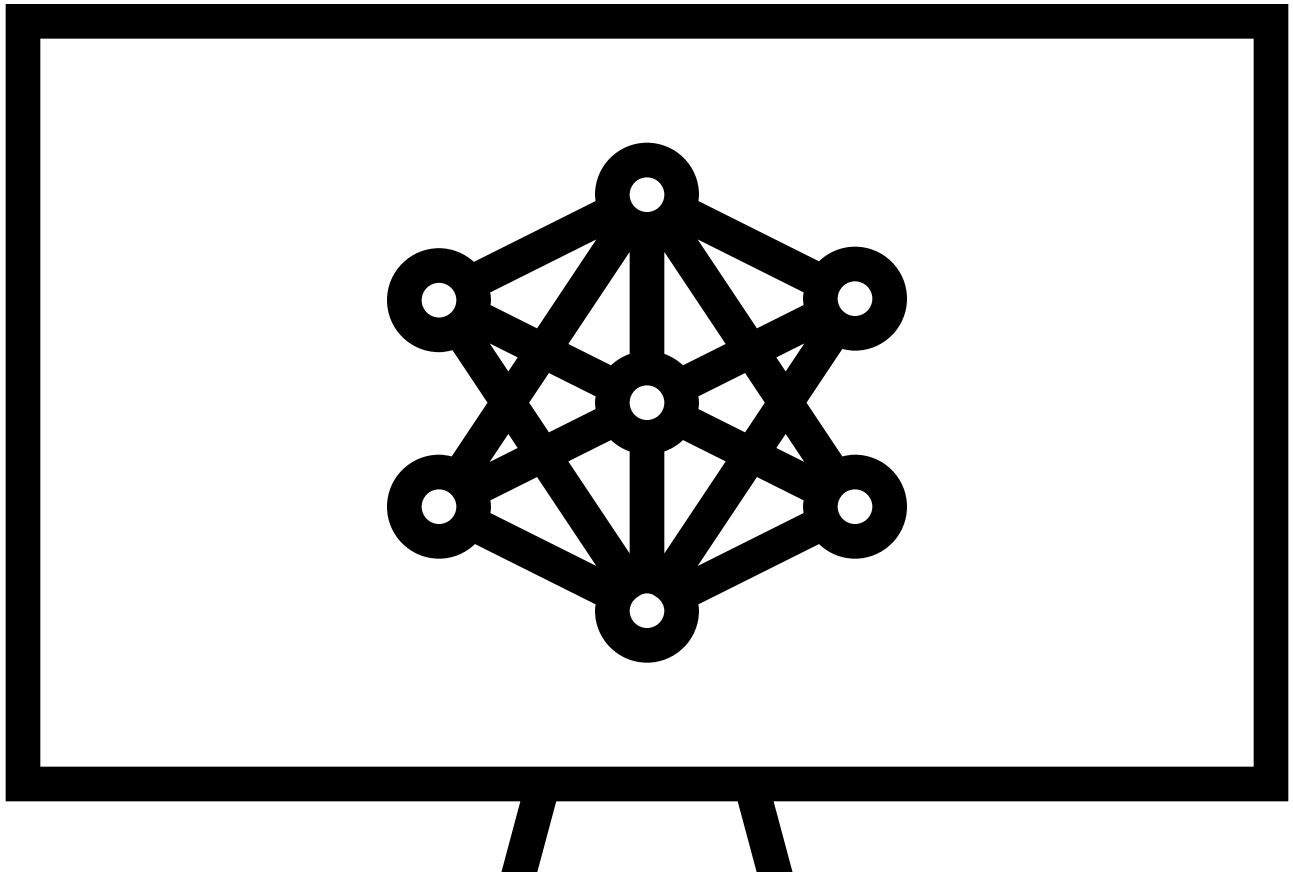
Use the right tools and technologies to safeguard AI applications with NeMo Guardrails scalable AI guardrail orchestration solution.



Access

To use the latest features and source code for adding AI guardrails to LLM applications, NeMo Guardrails is available as an open-source project on GitHub.

Access SDK



Try

Try the Nemotron Safety Guard models for content safety, topic control, and jailbreak detection on Hugging Face.

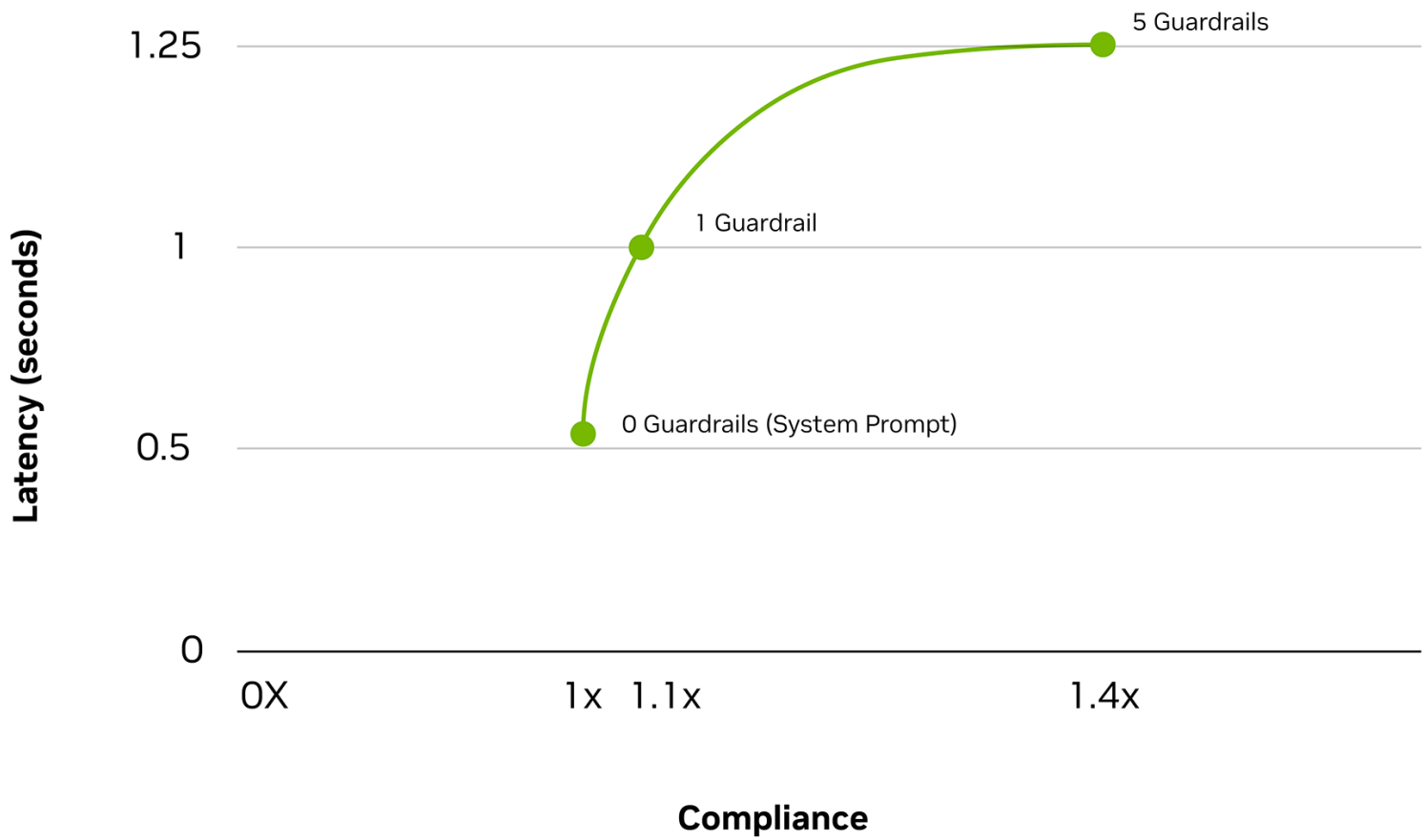
Try the Models

Performance

NeMo Guardrails enables AI guardrails to ensure that LLM responses are safe, secure, and compliant. Experience up to 1.4x improvement in compliance rate with a mere half-second of latency. Keep Enterprise AI operations safe and reliable by enforcing custom rules for AI models, agents, and systems. Use prepackaged NVIDIA NIM microservices that are optimized to make it easier to deploy.

Experience Over 1.4x Improvement in Compliance Rate
With Only Half a Second of Latency With NeMo Guardrails

Evaluated Policy Compliance With 5 AI Guardrails



The benchmark shows that orchestrating up to five GPU-accelerated guardrails in parallel with NeMo Guardrails increases policy compliance by 1.5x while adding only ~0.5 seconds of latency—delivering ~50% better protection without slowing down responses.

Starter Kits

Hero-Workflow

Safeguard your deployments with NemoGuard NIM microservices.

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Guardrails for RAG

Enhance content safety with programmable guardrails while building RAG apps delivering context-aware responses from vast multimodal enterprise data sources.

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Guardrails Evaluation

Measure the effectiveness and performance of AI guardrails in generative AI applications with an evaluation tool through NeMo Guardrails.

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NVIDIA NeMo Guardrails Learning Library

More Resources

Ethical AI

NVIDIA's platforms and application frameworks enable developers to build a wide array of AI applications. Consider potential algorithmic bias when choosing or creating the models being deployed. Work with the model's developer to ensure that it meets the requirements for the relevant industry and use case; that the necessary instructions and documentation are provided to understand error rates, confidence intervals, and results; and that the model is being used under the conditions and in the manner intended.